

Logistic Regression Example

Grace Carmichael

Department of Statistical Sciences
University of Cape Town

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Credit Card Default Data

A simulated data set containing information on ten thousand customers. The aim here is to predict which customers will default on their credit card debt.

default A binary variable indicating whether the customer defaulted on their debt ("No" or "Yes")

student A binary variable indicating whether a customer is a student ("No" or "Yes")

balance The average balance that the customer has remaining on their credit card after making their monthly payment

income Income of customer

Logistic Regression Output

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Call:
glm(formula = default ~ student + balance + income, family = "binomial",
    data = Default)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.4691  -0.1418  -0.0557  -0.0203   3.7383

Coefficients:
            Estimate      Std. Error z value      Pr(>|z|)
(Intercept) -10.869045196    0.492255516 -22.080 < 0.0000000000000002 ***
studentYes   -0.646775807    0.236252529  -2.738    0.00619 **
balance       0.005736505    0.000231895   24.738 < 0.0000000000000002 ***
income       0.000003033    0.000008203    0.370    0.71152

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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2920.6  on 9999  degrees of freedom
Residual deviance: 1571.5  on 9996  degrees of freedom
AIC: 1579.5

Number of Fisher Scoring iterations: 8
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$$\log\left(\frac{\widehat{p_i}}{1 - \widehat{p_i}}\right) = \hat{\beta}_0 + \hat{\beta}_1(\text{student}) + \hat{\beta}_2(\text{balance}) + \hat{\beta}_3(\text{income})$$

$$\log\left(\frac{\widehat{p_i}}{1 - \widehat{p_i}}\right) = -10.869 - 0.647(\text{student}) + 0.006(\text{balance}) + 0.000003(\text{income})$$

Interpreting β coefficients

$$\frac{\widehat{p_i}}{1 - p_i} = e^{\hat{\beta}_0} \times e^{\hat{\beta}_1(\text{student})} \times e^{\hat{\beta}_2(\text{balance})} \times e^{\hat{\beta}_3(\text{income})}$$

- student** Yes The odds of defaulting change by a factor of $e^{-0.647} = 0.5236$ when an individual is a student compared to not, holding all else constant. When an individual is a student, the odds of defaulting decrease by $(1 - e^{-0.647}) \times 100 = 47.6\%$ compared to when they are not a student.
- balance** The odds of defaulting change by a factor of $e^{0.0057} = 1.006$ for each unit increase in an individual's balance, holding all else constant. The odds of defaulting increases by $(e^{0.0057} - 1) \times 100 = 0.57\%$ for a unit increase in an individual's balance.
- income** The odds of defaulting change by a factor of $e^{0.000003} = 1.000003$ for each unit increase in an individual's income, holding all else constant. The odds of defaulting increases by $(e^{0.000003} - 1) \times 100 = 0.0003\%$ for a unit increase in an individual's income.

Hypothesis tests for estimated β coefficients

$$\beta_1$$

$$H_0: \beta_1 = 0$$

$$H_a: \beta_1 \neq 0$$

$$\alpha = 0.05$$

$$zscore = -2.738$$

$$p\text{-value} = 0.006$$

$$\beta_2$$

$$H_0: \beta_2 = 0$$

$$H_a: \beta_2 \neq 0$$

$$\alpha = 0.05$$

$$zscore = 24.738$$

$$p\text{-value} = < 0.001$$

$$\beta_3$$

$$H_0: \beta_3 = 0$$

$$H_a: \beta_3 \neq 0$$

$$\alpha = 0.05$$

$$zscore = 0.370$$

$$p\text{-value} = 0.711$$

Conclusions

- β_1 We can reject the null hypothesis because the p-value is less than the chosen significance level ($\alpha = 0.05$). We thus conclude that there is a difference in the likelihood of defaulting between students and non-students.
- β_2 We can reject the null hypothesis because the p-value is less than the chosen significance level ($\alpha = 0.05$). We thus conclude that an individual's balance affects their likelihood to default.
- β_3 We fail to reject the null hypothesis because the p-value is greater than the chosen significance level ($\alpha = 0.05$). We thus conclude that there is no relationship between an individual's income and their likelihood to default.

95% Confidence Intervals

 β_1

$$CI = \hat{\beta}_1 \pm Z_{\frac{\alpha}{2}} \times se(\hat{\beta}_1)$$

$$CI = -0.6468 \pm 1.96 \times 0.2363$$

$$CI = [-1.110, -0.184]$$

 β_2

$$CI = \hat{\beta}_2 \pm Z_{\frac{\alpha}{2}} \times se(\hat{\beta}_2)$$

$$CI = 0.0057 \pm 1.96 \times 0.00023$$

$$CI = [0.005, 0.006]$$

 β_3

$$CI = \hat{\beta}_3 \pm Z_{\frac{\alpha}{2}} \times se(\hat{\beta}_3)$$

$$CI = 0.000003 \pm 1.96 \times 0.000008$$

$$CI = [-0.00001, 0.00002]$$

Prediction

Predicted log-odds:

$$\log \left(\frac{\widehat{p_i}}{1 - \widehat{p_i}} \right) = -10.869 - 0.647(\text{student}) + 0.006(\text{balance}) + 0.000003(\text{income})$$

$$\log \left(\frac{\widehat{p_i}}{1 - \widehat{p_i}} \right) = -10.869 - 0.647(0) + 0.006(2020) + 0.000003(48500) = 0.866$$

Predicted probabilities:

$$p_i = \frac{1}{1 + e^{-\hat{LO}}}$$
$$p_i = \frac{1}{1 + e^{-(0.866)}} = 0.704$$

Confusion matrix

		Observed		
		1	0	Total
Predicted	1	169 (a)	137 (b)	a+b
	0	164 (c)	9530 (d)	c+d
Total		a+c	b+d	a+b+c+d

Positive Predictive Value

$$PPV = \frac{a}{a+b}$$
$$PPV = \frac{169}{169+137} = 0.552$$

Negative Predictive Value

$$NPV = \frac{d}{c+d}$$
$$NPV = \frac{9530}{164+9530} = 0.983$$

Sensitivity

$$Sensitivity = \frac{a}{a+c}$$
$$Sensitivity = \frac{169}{169+164} = 0.508$$

Specificity

$$Specificity = \frac{d}{b+d}$$
$$Specificity = \frac{9530}{137+9530} = 0.986$$